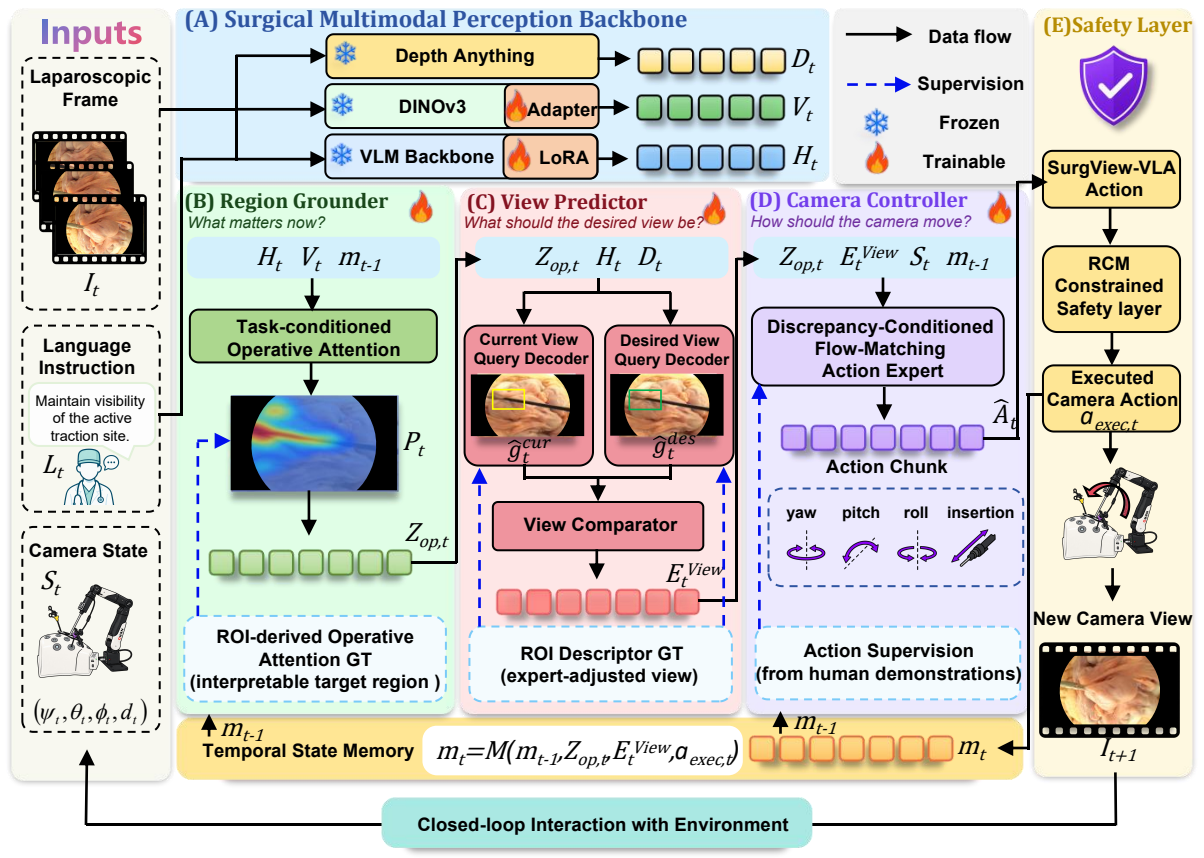

Inputs
Laparoscopic Frame
I_t
Language Instruction
Maintain visibility of the active traction site.
L_t
Camera State
S_t
$(\psi_t, \theta_t, \phi_t, d_t)$
(A) Surgical Multimodal Perception Backbone
Depth Anything
D_t
DINOv3
Adapter
V_t
VLM Backbone
LoRA
H_t
Data flow
Supervision
Frozen
Trainable
(E)Safety Layer
(B) Region Grounder
What matters now?
H_t V_t m_{t-1}
Task-conditioned Operative Attention
P_t
$Z_{op,t}$
ROI-derived Operative Attention GT (interpretable target region)
(C) View Predictor
What should the desired view be?
$Z_{op,t}$ H_t D_t
Current View Query Decoder
Desired View Query Decoder
$\hat{g}_t^{cur}$
$\hat{g}_t^{des}$
View Comparator
E_t^{View}
ROI Descriptor GT (expert-adjusted view)
(D) Camera Controller
How should the camera move?
$Z_{op,t}$ E_t^{View} S_t m_{t-1}
Discrepancy-Conditioned Flow-Matching Action Expert
$\hat{A}_t$
Action Chunk
yaw
pitch
roll
insertion
Action Supervision (from human demonstrations)
SurgView-VLA Action
RCM Constrained Safety layer
Executed Camera Action $a_{exec,t}$
New Camera View
I_{t+1}
m_{t-1}
Temporal State Memory
$m_t = M(m_{t-1}, Z_{op,t}, E_t^{View}, a_{exec,t})$
m_t
Closed-loop Interaction with Environment